

Keith's Notes for MA1200 PALSI: Precalculus

This document is primarily intended for the uses of students registered for PALSI at CityUHK, and participated in Keith TSE's section.

1 Introduction

These notes cover the key topics from MA1200 Calculus and Basic Linear Algebra I, including sets and functions, coordinate geometry and conic sections, polynomials and rational functions, trigonometric functions, and exponential and logarithmic functions.

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2 Sets and Functions

2.1 Set Notation

A set A is a collection of distinct objects (elements). Examples:

$\hat{A} = 1, 3, 5, 7, 9$ (odd numbers between 1 and 10).

$B = 1, 2, 3, 4, 5, \dots$ (positive integers).

$C = 0, \pm 3, \pm 6, \pm 9, \dots$ (multiples of 3).

$D = \mathbb{R}$ (real numbers).

Mathematical notation:

$p \in B$ means p belongs to B .

$p \notin B$ means p does not belong to B .

For example, if $E = 2, 3, 4, 5$, then $3 \in E$ and $\sqrt{6} \notin E$.

2.2 Equality and Subsets

Two sets A, B are equal ($A = B$) if they contain the same elements. E.g., $1, 2, 3 = 1, 2, 3$, but $1, 3, 4 \neq 1, 2, 3$.

$A \subseteq B$ if every element in A is in B . E.g., $1, 3 \subseteq 0, 1, 3, 4$, but $2, 4 \not\subseteq 0, 1, 3, 4$.

2.3 Functions

A function $f : X \rightarrow Y$ assigns each element in domain X to exactly one element in codomain Y . The range is the set of actual outputs. Examples of functions and their properties will be explored in subsequent sections.

3 Coordinate Geometry and Conic Sections

3.1 Coordinate Systems in 2D

Two systems: Cartesian and Polar.

3.1.1 Cartesian Coordinates

Position of point P given by (a, b) , where a is x-coordinate (distance from y-axis), b is y-coordinate (distance from x-axis).

Distance between $P_1 = (x_1, y_1)$, $P_2 = (x_2, y_2)$: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

Slope of line: $m = \frac{y_2 - y_1}{x_2 - x_1}$. Equation of line: Point-slope form $y - y_1 = m(x - x_1)$; slope-intercept $y = mx + c$.

Perpendicular lines have slopes $m_1 m_2 = -1$.

3.1.2 Polar Coordinates

Position by (r, θ) , $r \geq 0$, θ angle from positive x-axis (positive anti-clockwise, negative clockwise).

Relation to Cartesian: $x = r \cos \theta$, $y = r \sin \theta$, $r = \sqrt{x^2 + y^2}$, $\tan \theta = \frac{y}{x}$.

3.2 Conic Sections

Conics: circle, ellipse, parabola, hyperbola, from intersecting cone with plane.

3.2.1 Circle

Set of points equidistant from center $C = (h, k)$, radius r : $(x - h)^2 + (y - k)^2 = r^2$.

Standard: center $(0, 0)$, $x^2 + y^2 = r^2$.

Complete square to find center/radius.

3.2.2 Ellipse

Sum of distances to foci F_1, F_2 constant $2a$, $a > c > 0$, $b^2 = a^2 - c^2$.

Standard: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (foci on x-axis).

Vertices: $(\pm a, 0)$, $(0, \pm b)$.

General: $\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$.

If $a = b$, becomes circle.

Eccentricity $e = \frac{c}{a} < 1$.

3.2.3 Parabola

Equidistant from focus and directrix.

Standard: focus $(0, a)$, directrix $y = -a$: $x^2 = 4ay$.

General: vertex (h, k) , axis $y = k$: $(x - h)^2 = 4a(y - k)$; axis $x = h$: $(y - k)^2 = 4a(x - h)$.

3.2.4 Hyperbola

Difference of distances to foci constant $2a$, $c > a > 0$, $b^2 = c^2 - a^2$.

Standard: $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (foci on x-axis).

Vertices: $(\pm a, 0)$.

Asymptotes: $y = \pm \frac{b}{a}x$.

General: $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$.

Eccentricity $e = \frac{c}{a} > 1$.

3.3 Classification of Conics

General equation: $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$.

Discriminant $\Delta = B^2 - 4AC$:

$\Delta < 0$: ellipse (or circle if $A=C$, $B=0$).

$\Delta = 0$: parabola.

$\Delta > 0$: hyperbola.

If degenerate (e.g., $F=0$ in some cases), may be lines or point.

For rotation: If $B \neq 0$, rotate by $\theta = \frac{1}{2} \cot^{-1} \frac{A-C}{B}$ to eliminate xy -term.

Transformation: $x = x' \cos \theta - y' \sin \theta$, $y = x' \sin \theta + y' \cos \theta$.

4 Polynomials and Rational Functions

4.1 Polynomials

$P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, $n \geq 0$, $a_n \neq 0$.

Domain: $\mathbb{R}$.

Degree: n .

Examples: $x^5 - \sqrt{3}x^2 + 1$ (degree 5), constant 2 (degree 0).

Not polynomials: involve $\cos$, $\sqrt{x}$, negative exponents.

4.1.1 Quadratic Functions

$f(x) = ax^2 + bx + c$, $a \neq 0$.

Standard form: $f(x) = a(x-h)^2 + k$, vertex $(h, k) = \left(\frac{-b}{2a}, \frac{4ac-b^2}{4a}\right)$.

Graph: parabola, opens up if $a > 0$, down if $a < 0$. Range: $[k, \infty)$ if $a > 0$, $(-\infty, k]$ if $a < 0$.

Roots: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, discriminant $D = b^2 - 4ac$ (>0 two real, $=0$ one, <0 none).

4.2 Factorization of Polynomials

Remainder theorem: $P(x)$ divided by $(x-c)$, remainder $P(c)$.

Factor theorem: $(x-c)$ factor iff $P(c)=0$. Use trial/error for integer roots, long division or synthetic division.

4.3 Rational Functions

$r(x) = \frac{p(x)}{q(x)}$, p, q polynomials, $q \neq 0$.

Proper if $\deg p < \deg q$, improper otherwise. For improper, divide: quotient +

Domain: x where $q(x) \neq 0$.

4.4 Partial Fractions

Decompose proper rational into sum.

If $q(x) = (x-c_1)(x-c_2)\dots$ distinct linear: $\frac{A_1}{x-c_1} + \frac{A_2}{x-c_2} + \dots$. Repeated $(x-c)^k$: $\frac{A_1}{x-c} + \frac{A_2}{(x-c)^2} + \dots + \frac{A_k}{(x-c)^k}$. Irreducible quadratic $(ax^2 + bx + c)$: $\frac{Dx+E}{ax^2+bx+c}$.

Solve coefficients by equating or substituting. Examples as in lectures.

5 Trigonometric Functions

5.1 Definitions

In right triangle: $\cos \theta = \frac{\text{adj}}{\text{hyp}}$, $\sin \theta = \frac{\text{opp}}{\text{hyp}}$, $\tan \theta = \frac{\text{opp}}{\text{adj}}$, $0 < \theta < 90^\circ$.

Extended: unit circle, $x = \cos \theta$, $y = \sin \theta$, $\tan \theta = \frac{y}{x}$. Special angles: Reciprocals: $\sec \theta = 1/\cos \theta$, $\csc \theta = 1/\sin \theta$,

Angle	30°	45°	60°
sin	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$
cos	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$
tan	$1/\sqrt{3}$	1	$\sqrt{3}$

$$\cot \theta = 1/\tan \theta.$$

5.2 Radian Measure

1 rad = arc length on unit circle.

$$360^\circ = 2\pi \text{ rad}, 1^\circ = \pi/180 \text{ rad}.$$

5.3 Graphs

sin x, cos x: period 2π , range $[-1,1]$.

tan x: period π , range $\mathbb{R}$, undefined at $(2k+1)\pi/2$.

5.4 Identities

Basic: $\tan \theta = \sin \theta / \cos \theta$, $1 + \tan^2 \theta = \sec^2 \theta$, $1 + \cot^2 \theta = \csc^2 \theta$.

Compound angle:

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\tan(A \pm B) = (\tan A \pm \tan B)/(1 \mp \tan A \tan B)$$

Double angle:

$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2\cos^2 \theta - 1 = 1 - 2\sin^2 \theta,$$

$$\sin 2\theta = 2\sin \theta \cos \theta.$$

Half angle:

$$\cos(\theta/2) = \pm \sqrt{(1 + \cos \theta)/2},$$

$$\sin(\theta/2) = \pm \sqrt{(1 - \cos \theta)/2}.$$

Sum-to-product:

$$\sin A + \sin B = 2\sin((A+B)/2)\cos((A-B)/2)$$

$$\sin A - \sin B = 2\cos((A+B)/2)\sin((A-B)/2)$$

$$\cos A + \cos B = 2\cos((A+B)/2)\cos((A-B)/2)$$

$$\cos A - \cos B = -2\sin((A+B)/2)\sin((A-B)/2)$$

Product-to-sum:

$$\cos A \cos B = (1/2)[\cos(A+B) + \cos(A-B)]$$

$$\sin A \sin B = (1/2)[\cos(A-B) - \cos(A+B)]$$

$$\sin A \cos B = (1/2)[\sin(A+B) + \sin(A-B)]$$

5.5 Inverse Trig Functions

$\arcsin x$: domain $[-1, 1]$, range $[-\pi/2, \pi/2]$.

$\arccos x$: domain $[-1, 1]$, range $[0, \pi]$.

$\arctan x$: domain $\mathbb{R}$, range $(-\pi/2, \pi/2)$.

Properties: $\sin(\arcsin x) = x$, etc., within ranges.

5.6 General Solutions

$\sin \theta = k$: $\theta = (-1)^n \arcsin k + n\pi$.

$\cos \theta = k$: $\theta = \pm \arccos k + 2n\pi$.

$\tan \theta = k$: $\theta = \arctan k + n\pi$.

6 Exponential and Logarithmic Functions

6.1 Exponential Functions

$f(x) = a^x$, $a > 0$, $a \neq 1$.

Domain: $\mathbb{R}$, range $(0, \infty)$.

If $a > 1$, increasing; $0 < a < 1$, decreasing.

Limits: as $x \rightarrow \infty$, $a^x \rightarrow \infty$ if $a > 1$, $\rightarrow 0$ if $a < 1$; opposite as $x \rightarrow -\infty$.

Operations: $a^x a^y = a^{x+y}$, $a^x / a^y = a^{x-y}$, $(a^x)^y = a^{xy}$, $a^{-x} = 1/a^x$. Natural exp: $e \approx 2.718$, e^x .

Hyperbolic: $\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$.

Identities: $\cosh^2 x - \sinh^2 x = 1$, $\cosh 2x = \cosh^2 x + \sinh^2 x$.

6.2 Logarithmic Functions

$\log_a x = y$ where $a^y = x$, $a > 0$, $a \neq 1$, $x > 0$.

Inverse of exp: $\log_a(a^x) = x$, $a^{\log_a x} = x$.

Common log ($a = 10$), natural ln ($a = e$).

Properties:

$$\log_a(xy) = \log_a x + \log_a y,$$

$$\log_a(x/y) = \log_a x - \log_a y,$$

$$\log_a(x^k) = k \log_a x.$$

Change base: $\log_a x = \log_b x / \log_b a$.

Graphs: domain $(0, \infty)$, range $\mathbb{R}$, increasing if $a > 1$, decreasing if $0 < a < 1$.

7 Additional Materials for Problem-Solving

For conics, eccentricity e : circle $e = 0$, ellipse $0 < e < 1$, parabola $e = 1$, hyperbola $e > 1$.

Trig applications: solving triangles, identities for simplification in calculus.

Binomial applications: approximations $(1+x)^n \approx 1+nx$ for small x .

Partial fractions useful for integration.